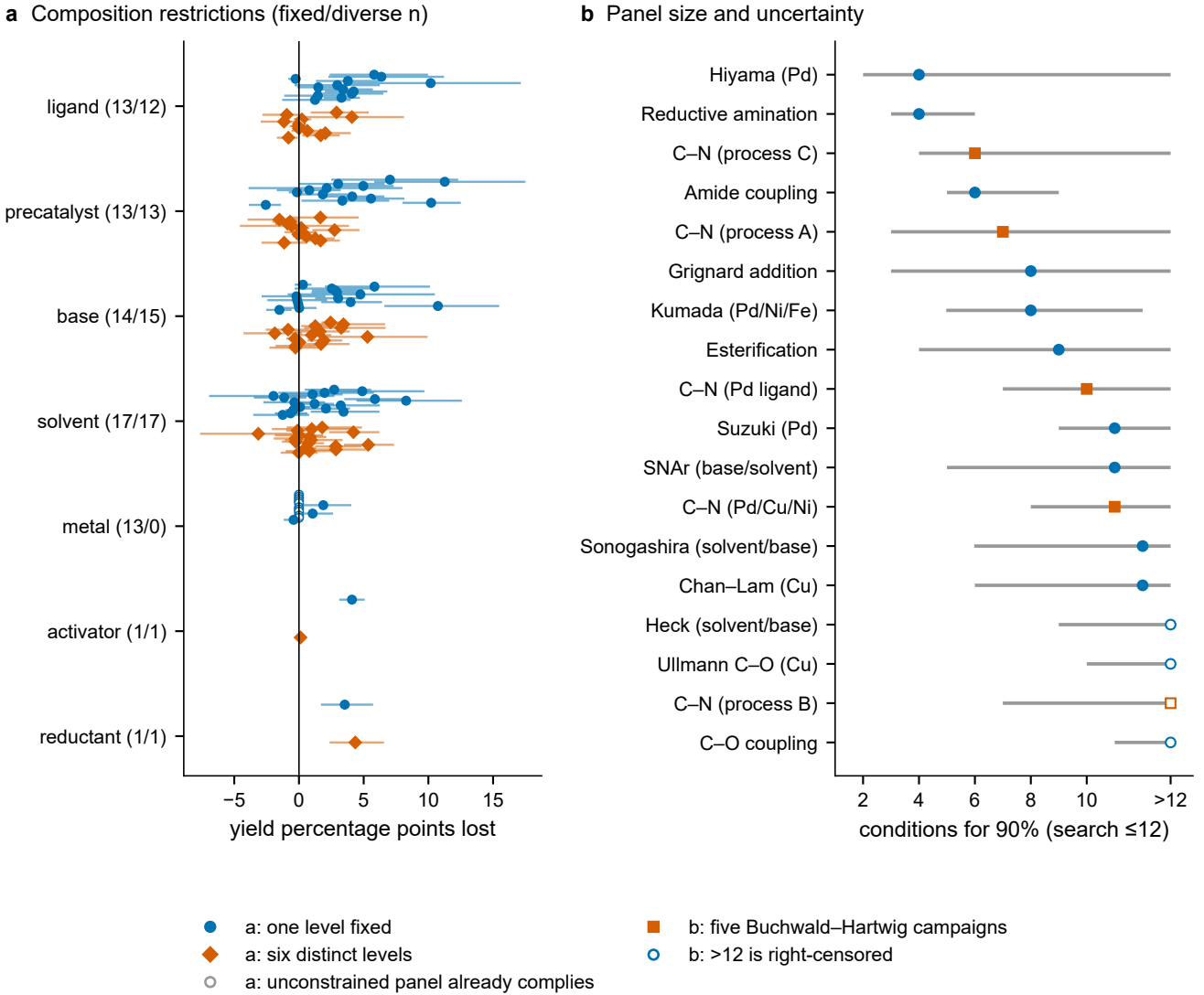

a Composition restrictions (fixed/diverse n)
ligand (13/12)
precatalyst (13/13)
base (14/15)
solvent (17/17)
metal (13/0)
activator (1/1)
reductant (1/1)
−5
0
5
10
15
yield percentage points lost
b Panel size and uncertainty
Hiyama (Pd)
Reductive amination
C–N (process C)
Amide coupling
C–N (process A)
Grignard addition
Kumada (Pd/Ni/Fe)
Esterification
C–N (Pd ligand)
Suzuki (Pd)
SNAr (base/solvent)
C–N (Pd/Cu/Ni)
Sonogashira (solvent/base)
Chan–Lam (Cu)
Heck (solvent/base)
Ullmann C–O (Cu)
C–N (process B)
C–O coupling
2
4
6
8
10
>12
conditions for 90% (search ≤12)
a: one level fixed
a: six distinct levels
a: unconstrained panel already complies
b: five Buchwald–Hartwig campaigns
b: >12 is right-censored